Oracle Utilities Service and Measurement Data Foundation v25.10HOT

Product Fix Design

Bug 38687121

Last Modified: January 21, 2026 7:13 AM
Printed On:

Oracle Utilities Service and Measurement Data Foundation Product Fix Design

Copyright 2026 by Oracle. All rights reserved.

The information contained in this document is proprietary and confidential to Oracle and is subject to the non-disclosure obligations described in the license agreement between your company and Oracle. You may not provide this document or any copy of this document to third parties outside your company unless you first obtain the written permission of Oracle.

This document introduces new and changed business and technical features that Oracle has included in the ”Single Fix” described in this document.

The functionality of the Single Fix described in this document may be superseded by functionality included in a future Single Fix, Release or Update. Oracle reserves the right to revise or expand the functionality of any future Single Fix, Release or Update of Oracle Utilities Service and Measurement Data Foundation in a manner compatible with Oracle’s business objectives, and the business objectives of each of Oracle’s customers, at the time that Oracle makes the software code generally commercially available.

Oracle will provide you with additional information concerning product Releases or Updates after Oracle finalizes the related specifications and tests the product component(s). Final documentation will be shipped to you with the media containing the Release or Update.

Please contact Oracle Customer Service Group if you have any questions or comments concerning this document, which has been prepared for your benefit as a licensee of Oracle Utilities Service and Measurement Data Foundation.

 Copyright ©2026, Oracle. All rights reserved.

Table of Contents

Table of Contents 3

Reported Issues 3
- Description 3
- Steps to Recreate 4

Solution Design 4
- Summary 5
- Changes 5

Recommended Test Cases 5
- Test Case 1: Verify that usage transaction is discarded when a measurement at start date/time is not found, scheduled read is not present and bill condition is initial. 5
- Test Case 2: Verify that usage transaction is discarded when it is not time for billing yet, even if there is a regular read present for that day. 7
- Test Case 3: Verify that get scalar rule is skipped for sub usage transaction and does not have scalar usage period SQ details when the scalar period is less than a day. 8

Reported Issues

Description

Billing is throwing an error if any new SA is created with the start date equal to expected work date of the schedule

Steps to Recreate

1. Create a device, device configuration, subtractive scalar measuring component.

Copyright ©2026, Oracle. All rights reserved.

2. Create a measurement cycle schedule that has the below expected work dates.
    - 04-15-2024
    - 05-15-2024
3. Create a service point with the above created measurement cycle and install the device on the service point.
4. Create readings on the scalar measuring component till 03-22-2024. There is no read on 04-15-2024.
5. Create usage calculation groups with get scalar usage calculation rules for both water service and sewer waste water service.
6. Create usage subscription for water and waste water services.
7. Sewer waste water usage subscription is stopped and a new one is created with start date as 04-15-2024.
8. The two usage subscriptions are as follows –
    - US 1 – active (water)
    - US 2 – activated on 04-15-2024 (sewer waste water)
9. On 04/15/2024, create usage transactions for US1 and US2. (with offsets (min – 5 and max – 1) and end range option as Derive Using Service Cycle Schedule and estimate date - 04-17-2024)
    - UT for US 1 - Start date/time : 03-12-2024 07:47AM PDT
    - End Date/Time – From : 04-15-2024 12:00AM PDT
    - End Date/Time – To : 04-15-2024 11:59PM PDT
    - UT for US 2 - Start date/time : 04-15-2024 12:00AM PDT, Bill condition – initial, automated retry : yes, provide retry until date/time
    - End Date/Time – From : 04-15-2024 12:00AM PDT
    - End Date/Time – To : 04-15-2024 11:59PM PDT
10. UT for US 1 moves to issue detected state with “Measurement not found for Measuring Component <> within the period from 04-09-2024 11:59PM PDT to 04-16-2024 11:59PM PDT. Not time to estimate yet.” error as it is not time to estimate.
11. Observe that UT for US2 moves to Approval in progress.

Solution Design

Changes are made to the Get Scalar Rule such that the following existing logic is only performed for sub usage transactions –

‘Do not execute the rule when usage transaction start date/time is on or after the Scalar Processing End Date From.’

Copyright ©2026, Oracle. All rights reserved.

Summary

Change Type	Y/N
Does this fix require any Admin (Soft) Table Changes to be made by the client?	N
Does this fix require upgrade data changes to be applied?	N
Does the fix involve DB Changes?	N
Is there an impact on any user interface or any other on-line behavior?	N
Is there an impact on any batch process?	N
Is there an impact on CM – Custom Modification processing?	N
Does this fix require an application viewer update for Algorithm data (run F1-AVALG)?	N
Does this fix require an application viewer update for MO data (run F1-AVMO)?	N
Does this fix require an application viewer update for Table data (run F1-AVTBL)?	N

Changes

Following java file has been modified –

GetScalarAlgComp_Impl.java

Recommended Test Cases

Test Case 1: Verify that usage transaction is discarded when a measurement at start date/time is not found, scheduled read is not present and bill condition is initial.

Pre-Setup

1. Create a device, device configuration, scalar measuring component.
2. Create a measurement cycle schedule that has the below expected work dates.
    - 04-15-2024
    - 05-15-2024

Copyright ©2026, Oracle. All rights reserved.

3. Create a service point with the above created measurement cycle and install the device on the service point.
4. Create readings on the scalar measuring component till 03-22-2024. There is no read on 04-15-2024.
5. Create usage calculation groups with get scalar usage calculation rules for both water service and sewer waste water service.
6. Create usage subscription for water and waste water services.
7. Sewer waste water usage subscription is stopped and a new one is created with start date as 04-15-2024.
8. The two usage subscriptions are as follows –
    - US 1 – active (water)
    - US 2 – activated on 04-15-2024 (sewer waste water)
9. On 04/15/2024, create usage transactions for US1 and US2. (with offsets (min – 5 and max – 1) and end range option as Derive Using Service Cycle Schedule and estimate date - 04-17-2024)
    - UT for US 1 - Start date/time : 03-12-2024 07:47AM PDT
    - End Date/Time – From : 04-15-2024 12:00AM PDT
    - End Date/Time – To : 04-15-2024 11:59PM PDT
    - UT for US 2 - Start date/time : 04-15-2024 12:00AM PDT, Bill condition – initial, automated retry : yes, provide retry until date/time (04-17-2024 11:59PM)
    - End Date/Time – From : 04-15-2024 12:00AM PDT
    - End Date/Time – To : 04-15-2024 11:59PM PDT
10. UT for US 1 moves to issue detected state with “Measurement not found for Measuring Component <> within the period from 04-09-2024 11:59PM PDT to 04-16-2024 11:59PM PDT. Not time to estimate yet.” error as it is not time to estimate.
11. UT for US 2 moves to discarded state with skip reason “No Schedule Read” and next schedule populated.

Case 1: When Waste water is processed first, then Water

12. On 04/17/2024, create a UT for US2 (waste water) with start date/time – 04-15-2024, end date/time from – 04-17-2024 12:00 AM, end date/time to – 04-17-2024 11:59PM, offsets (min – 5 and max – 1) and end range option as Derive Using Service Cycle Schedule, estimate date - 04-17-2024, Bill condition – initial, automated retry : yes, provide retry until date/time. (04-17-2024 11:59PM)
13. Verify that it is discarded with skip reason “No Scheduled Read” and next scheduled read date populated.
14. On 04/17/2024, create a UT for US1 (water) with start date/time – 03-12-2024 07:47 AM, end date/time from – 04-17-2024 12:00 AM, end date/time to – 04-17-2024 11:59PM, offsets (min – 5 and max – 1) and end range option as Derive Using Service Cycle Schedule, estimate date - 04-17-2024, automated retry : yes, provide retry until date/time. (04-17-2024 11:59PM)

 Copyright ©2026, Oracle. All rights reserved.

15. UT moves to approval in progress state and a read on 4/15/2024 gets estimated on the MC.

Case 2: When Water is processed first, then Waste Waster

16. On 04/17/2024, create a UT for US1 (water) with start date/time – 03-12-2024 07:47 AM, end date/time from – 04-17-2024 12:00 AM, end date/time to – 04-17-2024 11:59PM, offsets (min – 5 and max – 1) and end range option as Derive Using Service Cycle Schedule, estimate date - 04-17-2024, automated retry : yes, provide retry until date/time. (04-17-2024 11:59PM)
17. UT moves to approval in progress state and a read on 4/15/2024 gets estimated on the MC.
18. On 04/17/2024, create a UT for US2 (waste water) with start date/time – 04-15-2024, end date/time from – 04-17-2024 12:00 AM, end date/time to – 04-17-2024 11:59PM, offsets (min – 5 and max – 1) and end range option as Derive Using Service Cycle Schedule, estimate date - 04-17-2024, Bill condition – initial, automated retry : yes, provide retry until date/time. (04-17-2024 11:59PM)
19. Verify that it is discarded with skip reason “No Scheduled Read” and next scheduled read date populated.

Test Case 2: Verify that usage transaction is discarded when it is not time for billing yet, even if there is a regular read present for that day.

1. Create a device, device configuration, scalar measuring component.
2. Create a measurement cycle schedule that has the below expected work dates.
   - 06-12-2024
   - 07-12-2024
3. Create a service point with the above created measurement cycle and install the device on the service point.
4. Create readings on the scalar measuring component till 06-12-2024 07:45 AM. There exists a regular read on 06-12-2024 07:45 AM.
5. Create usage calculation group with get scalar usage calculation rule.
6. Create usage subscription with start date/time 06-12-2024 07:45 AM and with the above created usage calculation group.
7. Create a usage transaction with below details –
   - Scalar MC start date/time : 06-12-2024-00.00.00
   - Scalar MC End date/time from: 06-12-2024-00.00.00
   - Scalar MC End date/time to: 06-12-2024-23.59

Copyright ©2026, Oracle. All rights reserved.

- Bill condition - initial
- offsets (min – 5 and max – 1) and end range option as Derive Using Service Cycle Schedule and estimate date - 06-14-2024.
- Retry until date/time: 2024-06-14-23.59.59
- Automated retry: yes

8. Verify that it is discarded with skip reason “No Scheduled Read” and next scheduled read date populated.

Test Case 3: Verify that get scalar rule is skipped for sub usage transaction and does not have scalar usage period SQ details when the scalar period is less than a day.

1. Create a device, device configuration, scalar measuring component.
2. Create a service point and install the device on the service point.
3. Create measurements on the scalar measuring component from 01-01-2025 to 02-01-2025 12 A.M.
4. Create a usage calculation group with get scalar usage rule.
5. Add two rates in the D2-CCBRateScheduleLookup extendable lookup and configure the above created usage calculation group.
6. Create a usage subscription type with calculation group determination override algorithm - Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) configured. Ensure that D2-DRVUSGGRP soft parameter does not suppress one day usage.
7. Create a usage subscription using the above type and the rate history for the following dates –
   a. 01-01-2025 – Rate 1
   b. 02-01-2025 – Rate 2
8. Create a usage transaction for the following dates –

   Scalar Measuring component –
   Start Date/Time: 01-01-2025 12A.M
   End Date/Time From: 02-01-2025 12 A.M
   End Date/Time To: 02-01-2025 11:59 P.M
9. Verify that usage transaction moves to approval in progress and two sub usage transactions are created, 01-01-2025 to 02-01-2025 and 02-01-2025 to 02-05-2025. Sub UT 01-01-2025 to 02-01-2025 is in calculated state with SQ details.
10. Verify that sub usage transaction with dates 02-01-2025 and 02-05-2025 is in calculated state with No usage period SQ details. Get Scalar rule is not executed as start date/time

 Copyright ©2026, Oracle. All rights reserved.

– 02-01-2025 12 AM and scalar processing end date/time-from are same – 02-01-2025 12A.M (scalar period is less than one day)

Copyright ©2026, Oracle. All rights reserved.